



Six-state phase modulation for reduced crosstalk in a fiber optic gyroscope

CHUNXI ZHANG, SHAOBO ZHANG,* XIONG PAN, AND JING JIN

School of Instrumentation Science and Opto-Electronics Engineering, Beihang University, Beijing 100191, China

*zhangshaobo@buaa.edu.cn

Abstract: Electrical crosstalk in an interferometric fiber-optic gyroscope (IFOG) is regarded as the most significant factor influencing dead bands. Here, we present a six-state modulation (SSM) technique to reduce crosstalk. Compared to conventional four-state modulation (FSM) or square-wave modulation (SWM), the SSM reduces the correlation between modulation voltage and demodulation reference by separating their fundamental frequencies, and thus reduces the bias error induced by crosstalk. The measured dead band of a 1500-m IFOG is approximately 0.02 °/h using FSM and approximately 0.08 °/h using SWM, whereas there is no evidence of dead band using SSM. The IFOG using SSM also exhibits better angular random walk (ARW) and bias instability performance compared to the same IFOG using FSM or SWM. These results verify the crosstalk reduction effect of SSM. In theory, by using the relative intensity noise (RIN) suppressing technique with the optimal modulation depth of $2\pi/3$, the SSM can eliminate the crosstalk, which offers the potential for a high-performance IFOG with low noise, high sensitivity, wide dynamic range, and no dead band.

© 2018 Optical Society of America under the terms of the [OSA Open Access Publishing Agreement](#)

OCIS codes: (060.2370) Fiber optics sensors; (060.2800) Gyroscopes; (060.5060) Phase modulation.

References and links

1. H. C. Lefèvre, *The Fiber-Optic Gyroscope* (Artech House, 2014).
2. G. A. Sanders, S. J. Sanders, L. K. Strandjord, T. Qiu, J. Wu, M. Smiciklas, D. Mead, S. Mosor, A. Arrizon, W. Ho, and M. Salit, "Fiber optic gyro development at Honeywell," *Proc. SPIE* **9852**, 985207 (2016).
3. I. R. Edu, R. Obreja, and T. L. Grigorie, "Current technologies and trends in the development of gyros used in navigation applications-a review," in *Proceedings of the 5th WSEAS International Conference on Communications and Information Technology*, (World Scientific and Engineering Academy and Society, 2011), pp. 63–68.
4. G. A. Pavlath, "Closed-loop fiber optic gyros," in *Fiber Optic Gyros: 20th Anniversary Conference*, (International Society for Optics and Photonics, 1996), pp. 46–60.
5. O. Çelikel and S. E. San, "Design details and characterization of all digital closed-loop interferometric fiber optic gyroscope with superluminescent light emitting diode," *Opt. Rev.* **16**(1), 35–43 (2009).
6. A. M. Kurbatov, "New methods to improve the performance of open and closed loop fiber-optic gyros," *Gyroscopy Navigation* **6**(3), 207–217 (2015).
7. A. M. Kurbatov and R. A. Kurbatov, "Methods of improving the accuracy of fiber-optic gyros," *Gyroscopy Navigation* **3**(2), 132–143 (2012).
8. F. Sun, L. Wang, G. Wang, and G. Liu, "Study on the drift of modulated phase in interference fiber optic gyroscope," *J. Comput. (Taipei)* **5**, 394–400 (2010).
9. K.-H. Chong, W.-S. Choi, and K.-T. Chong, "Analysis of dead zone sources in a closed-loop fiber optic gyroscope," *Appl. Opt.* **55**(1), 165–170 (2016).
10. H. Gu, Y. Huan, A. Wang, and J. Luan, "Real-time dynamic simulation of angular velocity and suppression of dead zone in IFOG," *Opt. Rev.* **22**(1), 39–45 (2015).
11. D. A. Egorov, R. O. Olekhovich, A. A. Untilov, A. S. Aleinik, G. B. Deineka, and V. E. Strigalev, "Study on dead zones of fiber-optic gyros," *Gyroscopy Navigation* **2**(4), 197–207 (2011).
12. P. A. Ward, "Interferometric fiber optic gyroscope with off-frequency modulation signals," 7,817,284. U.S. Patent. 2010 Oct 19.
13. J. Jing, T. Hai-Ting, P. Xiong, and S. Ning-Fang, "Electrical crosstalk-coupling measurement and analysis for digital closed loop fibre optic gyro," *Chin. Phys. B* **19**(3), 030701 (2010).
14. J. G. Mark, D. A. Tazartes, and A. Cordova, "Method and apparatus for overcoming cross-coupling in a fiber optic gyroscope employing overmodulation," 5,682,241. U.S. Patent. 1997 Oct 28.
15. G. Spahlinger, "Fiber optic Sagnac interferometer with digital phase ramp resetting via correlation-free demodulator control," 5,123,741. U.S. Patent. 1992 Jun 23.

16. G. Spahlinger, M. W. Kemmler, M. Ruf, M. A. Ribes, and S. Zimmermann, "Error compensation via signal correlation in high-precision closed-loop fiber optic gyros," in *SPIE's 1996 International Symposium on Optical Science, Engineering, and Instrumentation*, (International Society for Optics and Photonics, 1996), pp. 218–227.
17. C.-J. Chen, "Interferometric Fiber Optic Gyroscope Dead Band Suppression," *Appl. Phys. Express* **1**, 072501 (2008).
18. G. Sanders, R. Dankwort, L. Strandjord, and R. Bergh, "Fiber optic gyroscope with deadband error reduction," 5,999,304. U.S. Patent. 1999 Dec 7.
19. P. Lo and R. A. Kovacs, "Fiber optic gyroscope with reduced non-linearity at low angular rates," 5,684,591. U.S. Patent. 1997 Nov 4.
20. D. A. Tazartes and G. A. Pavlath, "Automatic gain control for fiber optic gyroscope deterministic control loops," 7,859,678. U.S. Patent. 2010 Dec 28.
21. J. D. Zook, D. Chen, and G. N. Otto, "Temperature dependence and model of the electro-optic effect in LiNbO₃," *Appl. Phys. Lett.* **11**(5), 159–161 (1967).
22. F. Guattari, S. Chouvin, C. Moluçon, and H. Lefèvre, "A simple optical technique to compensate for excess RIN in a fiber-optic gyroscope," in *2014 DGON Inertial Sensors and Systems*, (Institute of Systems Optimization, 2014), pp. 1–14.
23. C.-J. Chen and C. H. Lange, "System and method for reducing fiber optic gyroscope color noise," 7,167,250. U.S. Patent. 2007 Jan 23.
24. J. Chamoun and M. J. F. Dignonnet, "Pseudo-random-bit-sequence phase modulation for reduced errors in a fiber optic gyroscope," *Opt. Lett.* **41**(24), 5664–5667 (2016).
25. J. Chamoun and M. J. F. Dignonnet, "Aircraft-navigation-grade laser-driven FOG with Gaussian-noise phase modulation," *Opt. Lett.* **42**(8), 1600–1603 (2017).

1. Introduction

Interferometric fiber-optic gyroscopes (IFOGs) have been extensively used in inertial rotation-sensing applications because of their remarkable fully solid-state configuration, small dimensions, low cost, etc. Simply by adjusting the fiber length or fiber coil diameter, the IFOGs can be adapted to various performances from rate grade to strategic grade [1–3]. In general, for most high-performance applications, the closed-loop configuration is preferred to the open-loop configuration because of the former's significantly improved dynamic range and scale factor linearity. At present, the double closed-loop processing method, which employs a square-wave biasing modulation, a digital phase ramp, and a second servo loop to stabilize the gain of the modulation chain, is the most adopted detection technique [4,5]. However, this detection method continues to limit performance to an unacceptable degree for certain high-performance applications.

The first limitation impacting these systems derives from the combination of the square-wave modulation (SWM) and the second servo loop. Synchronized with the SWM, the second servo loop controls the modulation gain by detecting a misalignment signal generated at the moment of ramp reset. Because the frequency of these resets depends on the rotation rate, the second servo loop is less efficient at low rotation rates because of the long reset time, which destabilizes the scale factor [6,7]. To address this issue, four-state modulation (FSM) has been implemented to replace the SWM [1,8]. With 2π phase shifts in each modulation period, the FSM is able to generate misalignment signals in every modulation period. Consequently, the modulation gain control loop functions at a constant high frequency, and thus, the scale factor stability is improved.

Another performance limitation is associated with the electrical crosstalk between the modulation voltage and the photodetector output signal. Because of the high correlation between the SWM analog voltage and the corresponding demodulation reference, the crosstalk can produce a bias error in closed-loop processing, which could cause a dead band (or dead zone), a critical error in closed-loop IFOGs, particularly at low rotation rates [9–14]. The aforementioned FSM is not a practical solution to the crosstalk problem, because its modulation voltage is also strongly correlated with its demodulation reference. Two widely used modulation techniques have been proposed to suppress the dead band due to crosstalk. The first technique eliminates the correlation between the crosstalk-induced bias errors and demodulation reference by random (pseudo-random in practice) modulation [15,16], which in turn increases random noise and bias instability [17]. The second technique is the dithering

technique, which shifts the modulation phase and prevents it from getting locked at one constant level [4,9,18,19]. However, this greatly reduces the dynamic range of the IFOG, and additional techniques are required to strip the dithering-induced bias errors and to stabilize the gain of the IFOG control loop [17,20]. Thus, both the dithering technique and random modulation have been of limited application in high-performance IFOGs, because they sacrifice accuracy and other performance characteristics to suppress the dead band.

In this work, we present a new modulation technique to reduce the crosstalk effects in IFOGs without sacrificing other performances. Based on the FSM scheme, we increase the number of states in each phase modulation cycle from four to six and then obtain two types of six-state modulation (SSM) schemes: SSMA and SSMB. Using either SSMA or SSMB, the fundamental frequency of the modulation voltage is separated from the frequency of the demodulation reference, resulting in a reduced correlation between the crosstalk-induced bias error and the demodulation reference. To verify the effectiveness of the proposed technique, we tested the performance characteristics of a 1500-m IFOG using SSMA or SSMB, including angular random walk (ARW), bias instability, and dead band, and compared these with the corresponding characteristics of the same IFOG using FSM and SWM. The experimental results show that the IFOG using SSMA exhibits unacceptably high noise, indicating that the SSMA is impractical in practice. However, the IFOG using SSMB exhibits better performance in all of these characteristics. The SSMB removes the dead band of the IFOG, which is approximately 0.08 °/h using SWM and approximately 0.02 °/h using FSM. Furthermore, when the modulation depth is $2\pi/3$, the SSMB can completely eliminate the crosstalk in theory, paving the way to a high-performance IFOG without dead band.

2. Principle and analysis

2.1 Basic operation of double closed-loop IFOGs

In a double closed-loop IFOG configuration, shown in Fig. 1, the main control loop consists of a light source, coupler, phase modulator, fiber coil, photodetector with preamplifier, analog/digital converter (ADC), logic processor, digital/analog converter (DAC), and its output buffer amplifier. The light source, coupler, phase modulator, and fiber coil comprise the well-known Sagnac interferometer, which produces a nonreciprocal phase difference between a pair of counter-propagating light waves when the fiber coil is subject to a rotation. The phase difference ϕ_s is proportional to the rotation rate Ω with a proportionality constant $K_s = 2\pi LD/\lambda c$, where L is the coil length, D is the coil average diameter, λ is the average source wavelength, and c is the light speed in vacuum. The interference signal of the counter-propagating light waves falls upon the photodetector, the light intensity of which varies as a cosine function of the phase difference. The light intensity, carrying the rotation rate information, is converted into the output photocurrent, which is then amplified and detected by the subsequent processing electronics. For the purpose of increasing the detection sensitivity, a bias modulation signal (e.g., SWM) is produced by the logic processor along with the DAC and buffer amplifier, and is applied to the phase modulator to bias the phase difference at the larger slope of the cosine curve, as illustrated by the bias points $-\phi_m$ and ϕ_m in Fig. 2(a). The modulated phase is synchronously demodulated in the logic processor, where the detected rotation rate is then integrated to produce the feedback ramp signal to nullify the rotation-induced phase shift. Because the total phase shift is always nearly equal to zero for all constant rotation rates, this closed-loop operation extends the dynamic range and increases the scale factor stability and linearity.

However, the scale factor may be less stable in the presence of temperature or other environmental variations, primarily because of the temperature-dependent electro-optic coefficient of the phase modulator [1,21]. This problem can be overcome by using the second servo loop (the dashed-line processing electronics in Fig. 1) to control the gain of the phase modulation chain. When subjected to a temperature change, the second servo loop will detect the error signal according to the applied bias modulation and then adjust the reference voltage

of the first DAC or the gain of the buffer amplifier with a second DAC to compensate the variation of the phase modulator electro-optic coefficient.

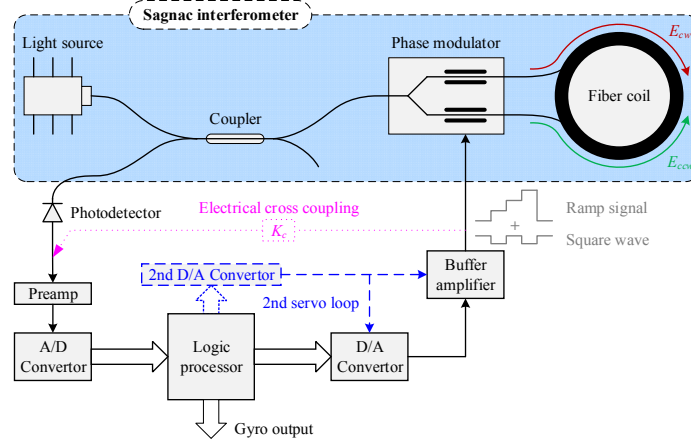


Fig. 1. Schematic of a double closed-loop IFOG showing the electrical cross-coupling path.

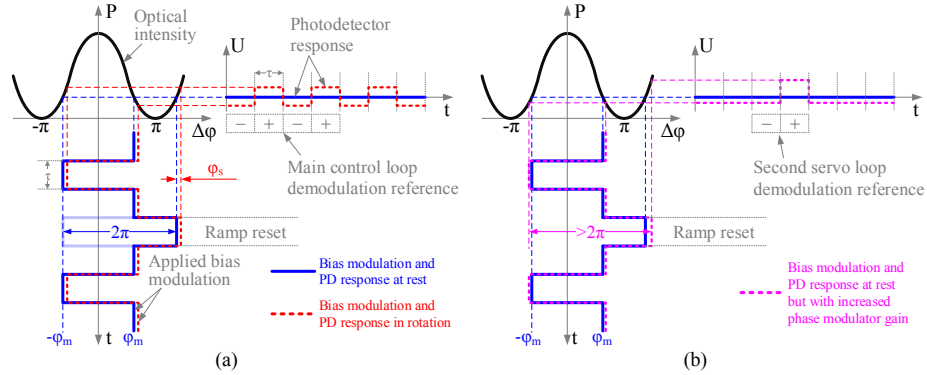


Fig. 2. Transfer function plot of an IFOG from phase shift to photodetector response (ignoring the transient spikes at each modulation state transition) with square-wave modulation (SWM) when subjected to (a) rotation rate and (b) increased phase modulator gain. The modulation depth is denoted by φ_m , and the Sagnac phase shift is denoted by φ_s . The corresponding demodulation reference is illustrated below each photodetector response. The phase shift undergoes a phase ramp reset at the fourth time interval τ , where τ is the transit time of light through fiber coil.

In the case of SWM, the phase modulator gain error can only be checked at each ramp reset, as depicted in Fig. 2(b), by comparing the sampled photodetector output signal just before and after the reset. Because the ramp reset frequency depends on the rotation rate, this second servo loop is activated unequally at different rotation rates, leading to an unstable and nonlinear scale factor. Furthermore, comparing Fig. 2(a) with 2(b), the second servo loop demodulation reference for checking the phase modulator gain error is the same as that used in the main control loop to demodulate the rate signal at the same time. That is, the main control loop and the second servo loop will interfere with each other [6], thus generating a series of spurious rate signals at and after the ramp reset moment.

To circumvent these problems, the FSM technique was designed specifically to improve the efficiency of the second servo loop. The FSM, in some sense, is obtained by superimposing a 2π phase shift modulation following the sequence $(-2\pi, 0, +2\pi, 0)$ or $(0, -2\pi, 0, +2\pi)$ with a state duration $\tau/2$ on the traditional SWM following the sequence of $(+\varphi_m, -\varphi_m)$ with a state duration τ (φ_m is the modulation depth, and τ is the transit time of light

through the fiber coil), yielding four modulation states $(-2\pi + \varphi_m, \varphi_m, 2\pi - \varphi_m, -\varphi_m)$ or $(\varphi_m, -2\pi + \varphi_m, -\varphi_m, 2\pi - \varphi_m)$, as shown in Figs. 3 and 4, respectively. Comparing Figs. 3(a) and 4(a) to 2(a), when subjected to a rotation, the two FSM schemes and the SWM yield the same photodetector responses: square wave signals (ignoring the transient spikes at each state transition) oscillating at the eigenfrequency $f_p = 1/(2\tau)$, and thus result in the same main control loop demodulation references: the sequence $(-1, +1)$ repeats at the eigenfrequency f_p . In contrast, the second servo loop demodulation references are different, as shown in Figs. 2(b), 3(b), and 4(b). Unlike the rate-dependent gain error detection process in SWM, both types of FSM schemes are able to yield a misalignment signal at each modulation half period τ , with the only difference being their opposite signs. These misalignment signals are detected with the demodulation sequence at $2f_p$, twice the eigenfrequency. As a result, the second servo loop functions at a constant high frequency and works independent of the main control loop. Because of the similarity of the two FSM schemes, only the one shown in Fig. 3 is used in the following analysis.

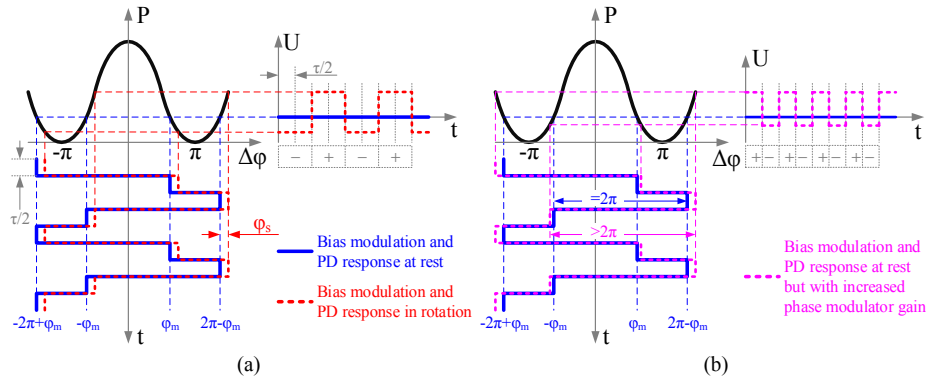


Fig. 3. Transfer function plot of an IFOG with the first four-state modulation (FSM) scheme, where the four sequential states are $-2\pi + \varphi_m, \varphi_m, 2\pi - \varphi_m$, and $-\varphi_m$, when subjected to (a) rotation rate and (b) increased phase modulator gain.

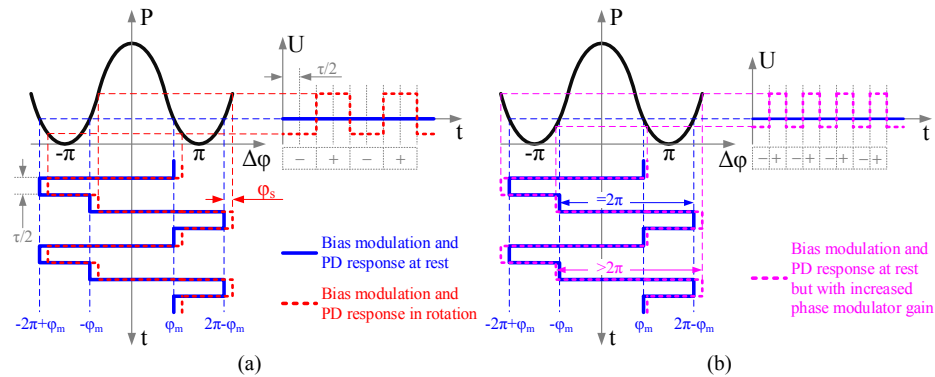


Fig. 4. Transfer function plot of an IFOG with the second FSM scheme, where the four sequential states are $\varphi_m, -2\pi + \varphi_m, -\varphi_m$, and $2\pi - \varphi_m$, when subjected to (a) rotation rate and (b) increased phase modulator gain.

2.2 Crosstalk effect analysis

The principal problem impacting the performance of double closed-loop IFOGs is crosstalk. Because electrical crosstalk is unavoidable in electronic designs, the applied modulation voltage can always couple into the extremely sensitive signal path of the photodetector. As stated above, the photodetector response containing the rotation rate information oscillates at

the eigenfrequency f_p , and is synchronously demodulated by the correlated sequence (with the same frequency and phase angle) to extract the rotation rate. Therefore, any interference signal correlated to the demodulation sequence can be demodulated into a bias error in the closed-loop processing, which would result in a dead band as well as an inaccurate rotation rate indication. Unfortunately, both the SWM and FSM analog voltages are highly correlated to the demodulation sequence, as will be examined below.

Because the Sagnac interferometer is operated as a differentiator [1], the bias modulation applied on the intensity-versus-phase mapping (as illustrated in Figs. 2 and 3) is the differential phase shift $\Delta\Phi_b(t)$ between the instantaneously modulated phase $\Phi_b(t)$ and the one delayed by time τ , $\Phi_b(t-\tau)$, which can be expressed as

$$\Delta\Phi_b(t) = \Phi_b(t) - \Phi_b(t-\tau), \quad (1)$$

and is further illustrated in Fig. 5. Accordingly, the modulated phase $\Phi_b(t)$ can be derived from an integral operation together with an insignificant initial value, as is shown for SWM and FSM in Fig. 5. Further, the modulated phase $\Phi_b(t)$ formed at the phase modulator is induced by the modulation analog voltage $V_b(t)$ driven by the DAC along with its buffer amplifier, and their relationship is formulated as

$$\Phi_b(t) = K_{pm} V_b(t), \quad (2)$$

where K_{pm} is the above-mentioned electro-optic coefficient of the phase modulator. The most crucial crosstalk is that between this modulation analog voltage $V_b(t)$ and the photodetector analog output, as shown in Fig. 1 [4]. Assuming the electrical coupling transfer function is a constant K_c , representing a linear system with zero phase shift, the interference error that is injected into the photodetector output is $V_e(t) = K_c V_b(t) = K_c \Phi_b(t) / K_{pm}$. Whether the interference error results in a spurious bias or random noise, and the size of the spurious bias, again depends on the correlation between $V_e(t)$ and the demodulation sequence $D(t)$ [16].

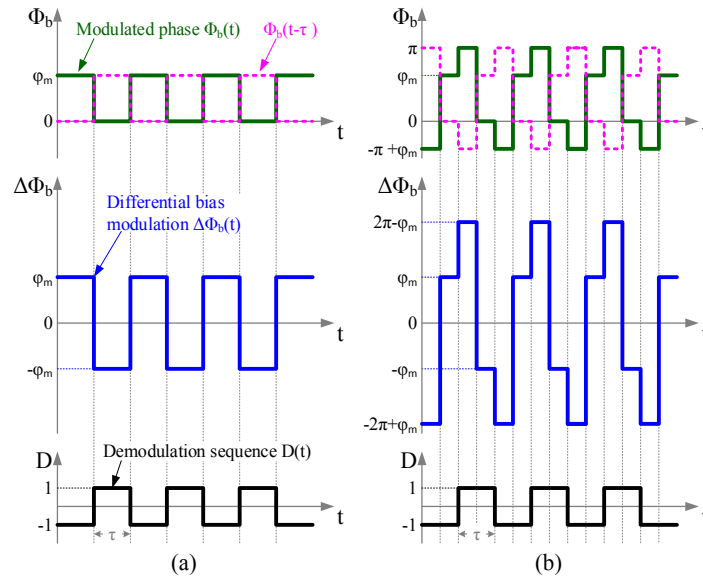


Fig. 5. Generation of the bias modulation $\Delta\Phi_b(t)$ by the difference between the modulated phase $\Phi_b(t)$ and $\Phi_b(t-\tau)$ with the (a) SWM and (b) FSM. The corresponding demodulation sequence $D(t)$ is illustrated below for the purpose of cross-correlation calculation.

For the sake of simplicity and comparison, the photodetector output is considered to be sampled at the frequency of $12f_p$ by the ADC, yielding 12 consecutive samples over each

demodulation cycle. Thus, the sampled interference error can be described as a periodic digital sequence, $E[n] = E[n + kN] = V_e(nT_s)$, where $n, k \in \mathbb{Z}$, N denotes the sequence period, equal to 12 here, and T_s denotes the sample period $1/(12f_p)$. In order to perform the synchronous demodulation, the corresponding demodulation sequence is also sampled synchronously and converted into the periodic digital sequence $D[n] = D[n + kN] = D(nT_s)$, similar in appearance to $E[n]$. The cross-correlation function of the two N -periodic sequence is defined as

$$\hat{R}_{ed}(k) = \frac{1}{N} \sum_{n=0}^{N-1} E[n] \cdot D[n-k], \quad (3)$$

where k is the correlation time lag. Owing to the assumed zero phase shift in the crosstalk transfer function, $k = 0$ is used to evaluate the size of the spurious bias φ_e , i.e.,

$$\varphi_e = G_{eq} \hat{R}_{ed}(0), \quad (4)$$

where G_{eq} denotes the equivalent gain for the closed-loop IFOG [16].

Considering the SWM in Fig. 5(a), we can derive the sampled interference error $E_{SWM}[n]$ and demodulation sequence $D_{SWM}[n]$, which are respectively presented as

$$E_{SWM}[n] = K_c \varphi_m / K_{pm} \cdot \{1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, \dots\} \quad (5)$$

and

$$D_{SWM}[n] = \{-1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, \dots\}. \quad (6)$$

Substituting the above into Eqs. (3) and (4), the spurious bias induced by the SWM voltage, φ_{eSWM} , can be calculated as:

$$\varphi_{eSWM} = \frac{G_{eq}}{N} \sum_{n=0}^{N-1} E_{SWM}[n] \cdot D_{SWM}[n] = -\frac{G_{eq} K_c}{2K_{pm}} \varphi_m = -K_e \varphi_m, \quad (7)$$

where $K_e = G_{eq} K_c / (2K_{pm})$ denotes the equivalent gain for the spurious bias. The above expression indicates that the absolute value of the spurious bias, φ_{eSWM} , is proportional to the modulation depth, φ_m , as is expected intuitively. Likewise, the spurious bias induced by the FSM voltage, φ_{eFSM} , can be calculated as:

$$\varphi_{eFSM} = \frac{G_{eq}}{N} \sum_{n=0}^{N-1} E_{FSM}[n] \cdot D_{FSM}[n] = K_e (\pi - \varphi_m), \quad (8)$$

where

$$E_{FSM}[n] = K_c / K_{pm} \cdot \{\varphi_m - \pi, \varphi_m - \pi, \varphi_m - \pi, \varphi_m, \varphi_m, \varphi_m, \pi, \pi, \pi, 0, 0, \dots\} \quad (9)$$

is the sampled interference error, and

$$D_{FSM}[n] = \{-1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, \dots\} \quad (10)$$

is the demodulation sequence, both of which are in accordance with Fig. 5(b). Comparing Eq. (8) with (7), the absolute value of the spurious bias, φ_{eFSM} , is also a function of φ_m , and $|\varphi_{eFSM}| = |\varphi_{eSWM}|$ when $\varphi_m = \pi/2$, which is the optimal value for the modulation depth, because it offers the maximum detection sensitivity [22].

Crosstalk-induced bias error that is constant during operation is not a serious problem, because it can easily be compensated by a software-based process. Unfortunately, because the modulation voltage $V_b(t)$ is superimposed on the ramp signal $V_r(t)$, and because the driving circuit is nonlinear and imperfect, the crosstalk-induced bias error φ_e will turn into a

feedback-voltage-dependent bias error $\varphi_e(V_r(t))$ [4,17,23]. In the closed-loop operation, when the input rotation rate is low, $\varphi_e(V_r(t))$ may get to a value $\varphi_e(V_r(t_0))$ that can nullify the Sagnac phase shift φ_s . Then the ramp signal $V_r(t)$ approaches constant and the closed-loop output signal gets blocked on zero [1]. When the input rotation rate is larger than the amplitude of $\varphi_e(V_r(t))$, the sum of $\varphi_e(V_r(t))$ and φ_s cannot be zero, and thus, the IFOG will not fall into a dead band [17]. That is, the dead band of an IFOG can be suppressed by reducing the amplitude of the feedback-voltage-dependent bias error $\varphi_e(V_r(t))$.

2.3 Reducing the crosstalk with the proposed six-state modulation (SSM) scheme

Apparently, reducing the crosstalk-induced bias error φ_e will reduce the amplitude of $\varphi_e(V_r(t))$, and thereby reduce the dead band. However, because the interference error occurs at the same frequency as the rotation rate information, the use of stop filters for interference in the signal path is inadvisable. In addition, the modulation voltage is generally within the range of a few volts, while the rotation rate signal is kept within the range of a few nanovolts for high-performance IFOGs, making it impractical to obtain the required high level of isolation on a common circuit board using electromagnetic compatibility measures [15]. The most effective approach to reducing the crosstalk-induced bias error is de-correlating the modulation voltage from the rotation rate signal, which is why the random modulation technique has been widely adopted [11]. The random modulation, however, increases random noise and bias instability because of the unstable optical power on the photodetector [17]. Furthermore, the use of pseudo-random sequences extends the demodulation period, which reduces the dynamic range and may induce a low-frequency disturbance that is hard to smooth.

In order to address the crosstalk issue without compromising other IFOG performance characteristics, we recently designed two types of modulation schemes, both of which theoretically exhibit not only the technical merit of the FSM, but also the ability to reduce the correlation between the modulation voltages and the rotation rate signals. The method of obtaining the two types of modulation schemes is similar to that of FSM. By superimposing a 2π phase shift modulation following the sequence $(0, -2\pi, 0, 0, +2\pi, 0)$ or $(0, 2\pi-2\varphi_m, 0, 0, 2\varphi_m-2\pi, 0)$ with a state duration $\tau/3$ on the traditional SWM following the sequence of $(+\varphi_m, -\varphi_m)$ with a state duration τ , we can obtain a modulation scheme with six modulation states $(\varphi_m, -2\pi + \varphi_m, \varphi_m, -\varphi_m, 2\pi - \varphi_m, -\varphi_m)$ or $(\varphi_m, 2\pi - \varphi_m, \varphi_m, -\varphi_m, -2\pi + \varphi_m, -\varphi_m)$, as shown in Figs. 6 and 7, respectively. For the convenience of the following discussion, the modulation scheme in Fig. 6 is called “six-state modulation A (SSMA)” and that in Fig. 7 is called “six-state modulation B (SSMB)”.

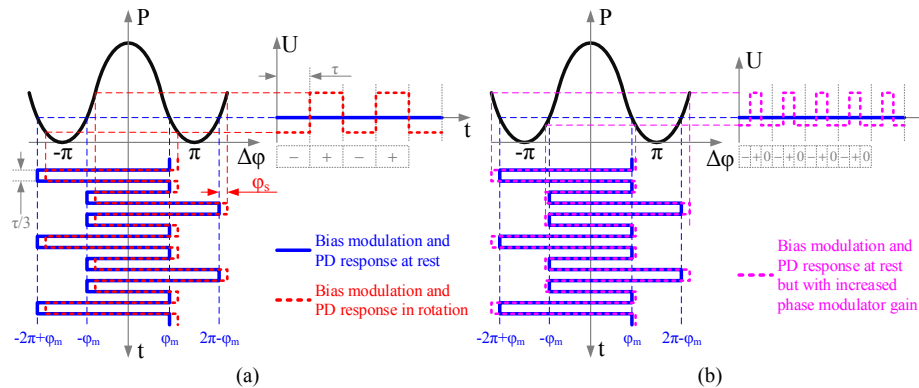


Fig. 6. Transfer function plot of an IFOG with six-state modulation A (SSMA), the six sequential states of which are $\varphi_m, -2\pi + \varphi_m, \varphi_m, -\varphi_m, 2\pi - \varphi_m$, and $-\varphi_m$, when subjected to (a) rotation rate and (b) increased phase modulator gain.

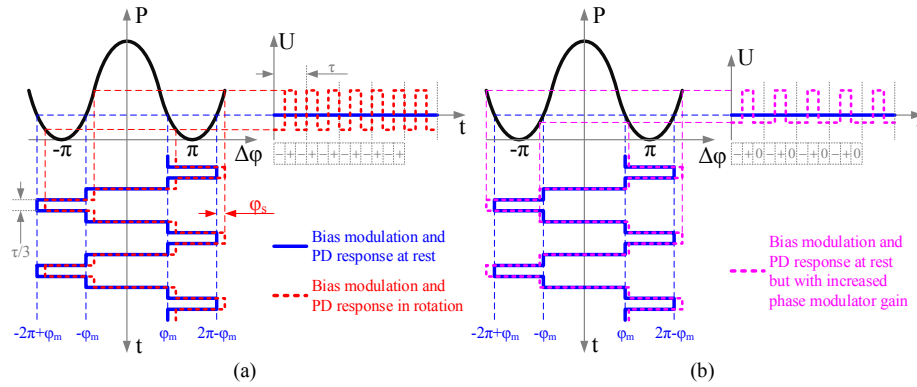


Fig. 7. Transfer function plot of an IFOG with six-state modulation B (SSMB), the six sequential states of which are ϕ_m , $2\pi-\phi_m$, ϕ_m , $-\phi_m$, $-2\pi+\phi_m$, and $-\phi_m$, when subjected to (a) rotation rate and (b) increased phase modulator gain.

As illustrated in Fig. 6(a), the SSMA photodetector response associated with a rotation rate remains unchanged from that of SWM (shown in Fig. 2(a)) and FSM (shown in Fig. 3(a)). The SSMA photodetector response induced by the phase modulator gain drift, as illustrated in Fig. 6(b), becomes a pulse train with the period of τ and width of $\tau/3$, which resembles the square wave response in Fig. 3(b). Clearly, the gain error of the phase modulator can be detected and compensated at every modulation half period τ , and the corresponding demodulation reference $(-1, +1, 0, -1, +1, 0, \dots)$ is orthogonal to that in the main control loop, $(-1, -1, -1, +1, +1, +1, \dots)$, indicating that the second servo loop works independent of the main control loop. Therefore, the SSMA possesses all the technical advantages of FSM, and as does the SSMB. Comparing Fig. 7(b) to 6(b), the phase modulator gain drift will cause exactly the same photodetector response in SSMB and SSMA. Further comparing Fig. 7(a) to 6(a), the rotation rate-induced photodetector response in SSMB is also a square wave, but with three times the frequency of that in SSMA. The demodulation reference in the main control loop, $(-1, +1, -1, +1, -1, +1, \dots)$, remains orthogonal to that in the second servo loop, $(-1, +1, 0, -1, +1, 0, \dots)$, confirming the above conclusion.

As with the FSM, the modulated phase $\Phi_b(t)$ of SSMA or SSMB, which is proportional to the modulation voltage $V_b(t)$, is graphed in Fig. 8(a) or 8(b), respectively, together with the corresponding demodulation sequence $D(t)$. Note that the peak-to-peak amplitude of $\Phi_b(t)$ in SSMA or SSMB is $2\pi-\phi_m$, the same value as in the case of FSM. Therefore, the driving circuit does not need to provide a larger output range, and the SSMA and SSMB will not increase the nonlinearity of the modulation chain, compared to the FSM.

Following the same analysis procedure explained in the previous section, the spurious biases induced by the SSMA and SSMB, φ_{eSSMA} and φ_{eSSMB} , can be calculated as follows

$$\varphi_{eSSMA} = \frac{G_{eq}}{N} \sum_{n=0}^{N-1} E_{SSMA}[n] \cdot D_{SSMA}[n] = K_e \left(\frac{2}{3} \pi - \phi_m \right), \quad (11)$$

where

$$E_{SSMA}[n] = K_c / K_{pm} \cdot \{\varphi_m, \varphi_m, \varphi_m - \pi, \varphi_m - \pi, \varphi_m, \varphi_m, 0, 0, \pi, \pi, 0, 0, \dots\} \quad (12)$$

is the sampled interference error of SSMA, and

$$D_{SSMA}[n] = \{-1, -1, -1, -1, -1, -1, +1, +1, +1, +1, +1, +1, \dots\} \quad (13)$$

is the sampled demodulation sequence of SSMA, as shown in Fig. 8(a); likewise,

$$\varphi_{eSSMB} = \frac{G_{eq}}{N} \sum_{n=0}^{N-1} E_{SSMB}[n] \cdot D_{SSMB}[n] = K_e \left(\frac{2}{3} \pi - \varphi_m \right), \quad (14)$$

where

$$E_{SSMB}[n] = K_c / K_{pm} \cdot \{\varphi_m, \varphi_m, \pi, \pi, \varphi_m, \varphi_m, 0, 0, \varphi_m - \pi, \varphi_m - \pi, 0, 0, \dots\} \quad (15)$$

is the sampled interference error of SSMB, and

$$D_{SSMB}[n] = \{-1, -1, +1, +1, -1, -1, +1, +1, -1, -1, +1, +1, \dots\} \quad (16)$$

is the sampled demodulation sequence of SSMB, as shown in Fig. 8(b). Comparing Eq. (11) with (14), the spurious biases induced by the SSMA and SSMB can be seen to be exactly the same. Comparing the result in Eq. (11) to that in Eq. (8), we can obtain

$$|\varphi_{eSSMA}| = \frac{1}{3} |\varphi_{eFSM}| \text{ when } \varphi_m = \pi/2, \quad (17)$$

indicating that the proposed SSMA and SSMB will cause smaller bias errors, as expected. This reduction of the crosstalk can be attributed to the separation of the fundamental frequency between the modulation voltage $V_b(t)$ and the demodulation reference $D(t)$. As shown in Fig. 8, the fundamental frequency of the SSMA modulation voltage is $3f_p$, three times higher than the frequency of the corresponding demodulation reference f_p , whereas the fundamental frequency of the SSMB modulation voltage is f_p , three times lower than the frequency of the corresponding demodulation reference $3f_p$.

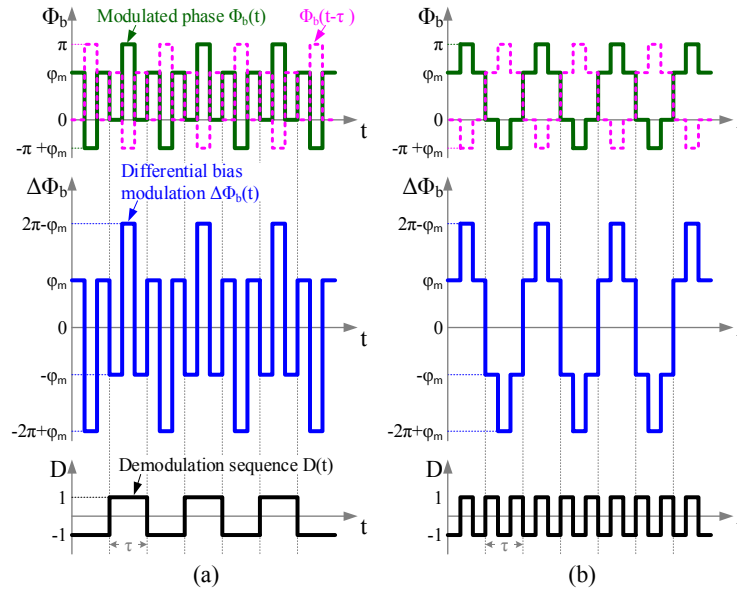


Fig. 8. Generation of the bias modulation $\Delta\Phi_b(t)$ by the difference between the modulated phase $\Phi_b(t)$ and $\Phi_b(t-\tau)$ with the (a) SSMA and (b) SSMB. The corresponding demodulation sequence $D(t)$ is illustrated below for the purpose of cross-correlation calculation.

For a high-performance IFOG, however, the best modulation depth is not $\pi/2$. For the purpose of increasing the signal-to-noise ratio (SNR) of the photodetector, the modulation depth is usually chosen to approach π to suppress the excess relative intensity noise (RIN) of the light source, the most fundamental noise contributor of the IFOG [1,22,24]. However, the compromise between noise and other performances, including sensitivity, dynamic range,

shock survival, scale factor stability, and thermal stability, demands an optimal modulation depth between $\pi/2$ and $7\pi/8$. Furthermore, with an improved RIN suppressing technique [22] or a laser-driven IFOG [24,25], the optimal modulation depth becomes closer to $\pi/2$.

For a more comprehensive comparison, Fig. 9 shows plots of $|\varphi_{eSWM}|$, $|\varphi_{eFSM}|$, and $|\varphi_{eSSMA}|$ (or $|\varphi_{eSSMB}|$) as normalized functions of modulation depth, with the typical optimal bias zone (from $\pi/2$ to $7\pi/8$) highlighted. With SWM, the absolute value of the induced bias error, $|\varphi_{eSWM}|$, increases monotonically with increasing modulation depth. Conversely, with FSM, $|\varphi_{eFSM}|$ decreases monotonically with increasing modulation depth. These values intersect at $\pi/2$, the starting point of the optimal bias zone, showing that the FSM exhibits better crosstalk reduction (the deeper the bias point, the better the reduction) than the SWM within the optimal bias zone. The $|\varphi_{eFSM}|$ vanishes when $\varphi_m = \pi$, but this is an inoperative bias point. However, in the case of SSM (SSMA or SSMB), the induced bias error also vanishes, but at $2\pi/3$, a bias point that is operative and inside the optimal bias zone. This is clearly an attractive advantage over SWM and FSM, for it offers the possibility of completely eliminating the crosstalk-induced bias error. Furthermore, even excluding $2\pi/3$, the SSM exhibits better crosstalk reduction than FSM in the most optimal bias zone, except in the fractional range from $5\pi/6$ to $7\pi/8$. In the exceptional bias zone, the FSM does not offer obvious advantages over SSM, as presented in Fig. 9, especially considering the above-mentioned technology trend, whereby the optimal modulation depth becomes closer to $\pi/2$.

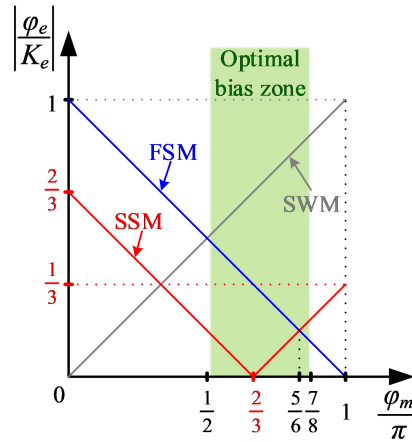


Fig. 9. Normalized absolute value of the crosstalk-induced bias errors versus modulation depth for different modulation schemes, with the optimal modulation depths highlighted.

As a practical matter, the crosstalk transfer function cannot be the assumed ideal constant K_c [9], and therefore the non-zero phase shift should be considered to estimate the bias error, because the corresponding correlation time lag k is a major determinant of the cross-correlation function $\hat{R}_{ed}(k)$ in Eq. (3). By scanning k , the normalized bias errors φ_e/K_e of SWM, FSM, SSMA, and SSMB (with the modulation depths of $\pi/2$, $2\pi/3$, $3\pi/4$, and $7\pi/8$, respectively) are plotted versus phase shift (in degrees) in Fig. 10. The phase scanning resolution is 1° and the scanning scope is 0° to 180° , which is sufficient to estimate the bias error, because the cross-correlation function is periodic at 360° and generally symmetric around the 180° -axis in phase.

First, considering only the $\pi/2$ modulation depth (solid lines in Fig. 10), the bias error in the SWM presents as a straight line crossing the 0° -axis at 90° phase shift, which is coincident with the theoretical prediction that the interference error $V_e(t)$ is orthogonal to the demodulation sequence $D(t)$ at the 90° phase shift. The maximum absolute value of the normalized bias error in SWM is 0.5, occurring at phase shifts of 0° or 180° . Conversely, the

bias error in the FSM exhibits a peak value of 1 at the 90° phase shift and always exceeds 0.5 within 120° of phase shift. In SSMA, the bias error is also equal to 0 at the 90° phase shift because of the orthogonality of $V_e(t)$ and $D(t)$; however, in this case, the bias error achieves peak values of 0.5 and -0.5 at phase shifts of 60° and 120° , respectively. Unlike the 360° periodicity of the bias error in SWM, FSM, or SSMA, the bias error in SSMB exhibits periodicity of 120° and symmetry around the 60° -axis, and equals to 0 at 30° of phase shift, all of which are coincident with the fact that the frequency of $D(t)$ in SSMB is three times higher than that in SWM, FSM, or SSMA. The maximum absolute value of the normalized bias error in SSMB, occurring at phase shifts of 0° or 60° , equals $1/6$, which is the smallest value among these modulation schemes. Furthermore, at the same phase shift, SSMB also exhibits the smallest bias error among all of the modulation schemes.

At the modulation depths of $2\pi/3$, $3\pi/4$, and $7\pi/8$, the corresponding bias errors of SWM, FSM, or SSMA behave in the same manner as the bias error at the $\pi/2$ modulation depth when rotated counterclockwise by various angles (the larger the modulation depth, the larger the rotation angle) around the point at 90° , or around 30° in the case of the SSMB. Thus, the area (shaded in Fig. 10) enclosed by the bias error's upper boundary (corresponding to the $\pi/2$ modulation depth) and lower boundary (corresponding to the $7\pi/8$ modulation depth) of each modulation scheme contains all the possible induced bias errors within the optimal bias zone. These shaded areas in Fig. 10 show that, at any given phase shift and modulation depth, the bias error in SSMB is the smallest in most cases. The bias errors in SSMA are exactly the same as those in SSMB at zero phase shift (in accordance with Fig. 9), but diverge at non-zero phase shifts. Most remarkably, at the modulation depth of $2\pi/3$, the bias error in SSMB always equals zero regardless of the phase shift. Consequently, the SSMB offers significant promise as a modulation scheme for reducing the crosstalk effects in an IFOG, especially at the modulation depth of $2\pi/3$.

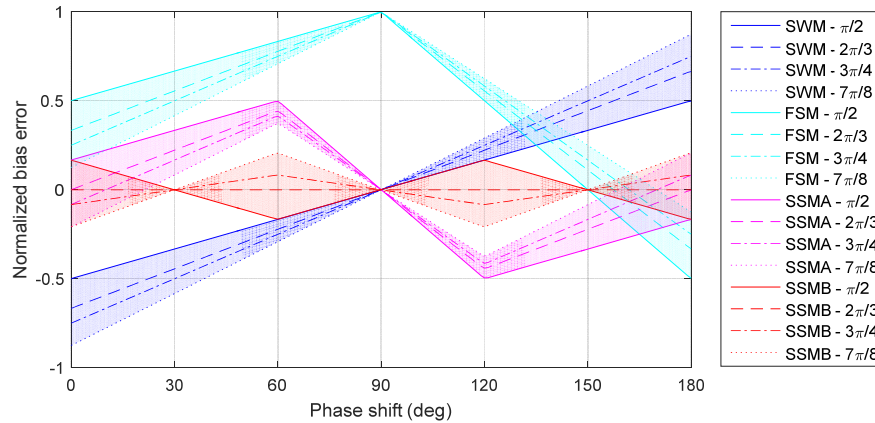


Fig. 10. Normalized bias errors induced by crosstalk versus phase shift of the coupling transfer function for different modulation schemes with different modulation depths.

3. Experiments and discussion

An IFOG with the arrangement shown in Fig. 1 was assembled and tested for dead band to examine the crosstalk reduction effects of the proposed SSM schemes. The light source was an amplified spontaneous emission broadband source with a 1550-nm central wavelength. The fiber coil used was nominally 1500 meters of polarization-maintaining fiber with a nominal diameter of 100 mm, making the transit time roughly equal to $7.5 \mu\text{s}$. In the absence of any RIN suppressing technique, the IFOG noise would decrease with increasing modulation depth, and $\pi/2$, $2\pi/3$, and $7\pi/8$ were adopted to verify the previous analysis.

Before the dead band testing, the noise and drift performances were measured and compared to verify the practicability of the proposed modulation schemes. Figure 11 illustrates the measured Allan deviation of the IFOG outputs driven by the four modulation schemes (SWM, FSM, SSMA, and SSMB) with modulation depths of $\pi/2$, $2\pi/3$, and $7\pi/8$, and the corresponding ARWs and bias instabilities are summarized in Table 1. The results show that the ARW and the bias instability of SSMA at any of the modulation depths were more than three times larger than those of other modulation schemes. This is probably because the SSMA modulation voltage contains more high-frequency components, which would increase the noise and interference on the electronic circuit board. Consequently, the SSMA was demonstrated to be impractical and was not considered in the subsequent experiments. In the case of SSMB, however, the measured ARW and bias instability performance were better than those in the SWM and FSM. This improvement was increasingly obvious as the modulation depth increased. Because the noise generally increases or decreases along with the crosstalk, the improvement in ARW and bias instability performance was reasonable, and the crosstalk reduction effect of SSMB was verified from another perspective.

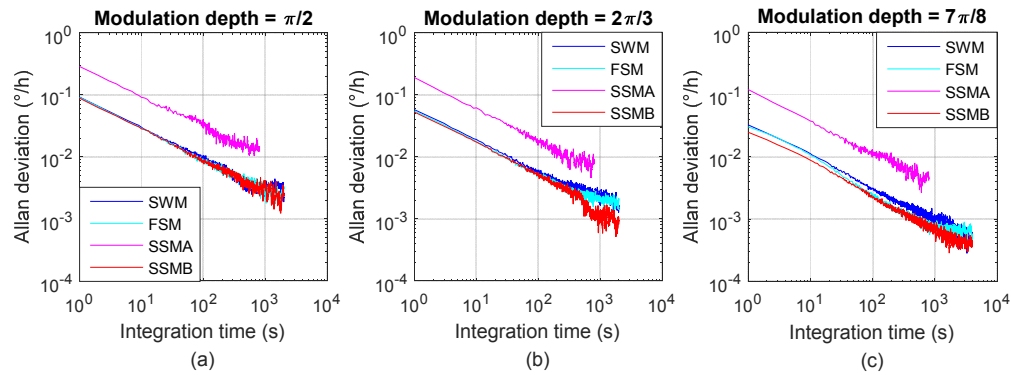


Fig. 11. Measured Allan deviation of the IFOG outputs for different modulation schemes with different modulation depths: (a) $\pi/2$, (b) $2\pi/3$, (c) $7\pi/8$.

Table 1. Evaluated ARW and bias instability of the IFOG according to the Allan deviation in Fig. 11

Modulation depth	ARW ($^{\circ}/\sqrt{h}$)				Bias instability ($^{\circ}/h$)			
	SSMA	SWM	FSM	SSMB	SSMA	SWM	FSM	SSMB
$\pi/2$	0.0048	0.0015	0.0015	0.0015	0.0150	0.0030	0.0020	0.0020
$2\pi/3$	0.0032	0.0010	0.0009	0.0008	0.0090	0.0020	0.0020	0.0010
$7\pi/8$	0.0020	0.0005	0.0004	0.0003	0.0060	0.0010	0.0007	0.0005

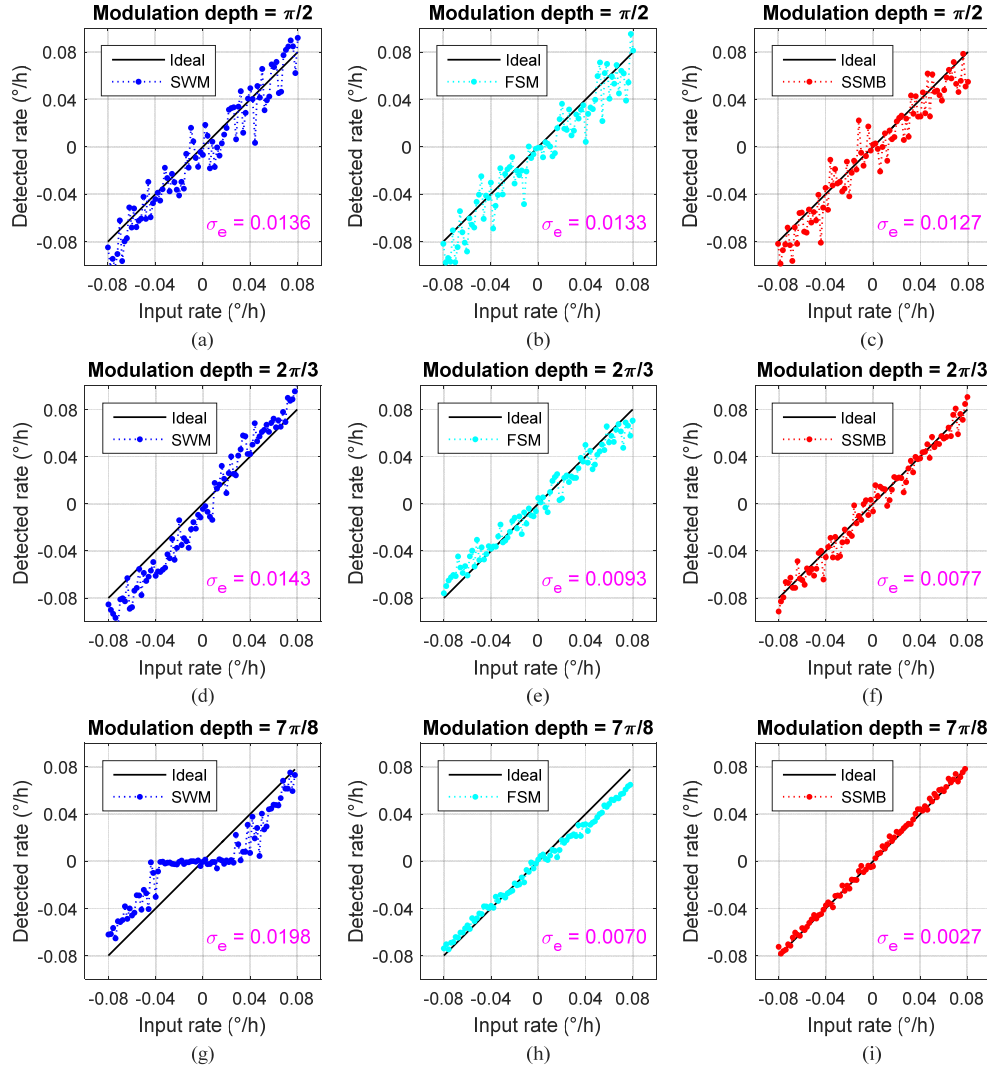


Fig. 12. Dead band measurement results of the IFOG for different modulation schemes: [(a), (d), (g)] the SWM, [(b), (e), (h)] FSM, and [(c), (f), (i)] SSMB, with different modulation depths: [(a), (b), (c)] $\pi/2$, [(d), (e), (f)] $2\pi/3$, and [(g), (h), (i)] $7\pi/8$. The standard deviation, σ_e , of each IFOG output from the ideal rate is presented at bottom right of each plot.

To test the dead band, the IFOG was mounted on a turntable with the sensitive axis perpendicular to the turntable axis. By pointing the IFOG's sensitive axis east and rotating the turntable at a very low speed of 0.0001 °/s (corresponding to a rate step of 0.00002 °/h for each second), the IFOG scanned through a small fraction of the earth rate that ranged from approximately -0.08 °/h to 0.08 °/h. The measured data were averaged over 100 s to reduce the noise, leading to the input rate resolution of 0.002 °/h. The IFOG outputs versus rotation rate are separately plotted for different modulation schemes and modulation depths in Fig. 12. Because of the low SNR of the IFOG at modulation depths of $\pi/2$ and $2\pi/3$, it is difficult to distinguish the dead band in Figs. 12(a) to 12(f). At the modulation depth of $7\pi/8$, however, the figure clearly shows that the dead band was approximately 0.08 °/h with SWM (Fig. 12(g)) and approximately 0.02 °/h with FSM (Fig. 12(h)), whereas no evidence of dead band was observed with SSMB (Fig. 12(i)). This result agrees well with the previous prediction (in Fig. 10) when there was a small phase shift applied to the interference error.

In addition, for the points outside the dead band, the detected rate of SWM (Fig. 12(g)) and FSM (Fig. 12(h)) diverged markedly from the ideal rate, reflecting the nonlinearity of the scale factor as well as the dead band. To evaluate the crosstalk-induced bias errors in Figs. 12(a) to 12(f), the standard deviation from the ideal rate of each experiment was calculated and presented at the bottom right of each plot. Comparing the standard deviations of the same modulation scheme with different modulation depths, the standard deviation of the FSM or SSMB decreased with increasing modulation depth, as a result of the reduction of RIN combined with some reduction of crosstalk. In contrast, in spite of the reduction of RIN, the standard deviation of the SWM increased with increasing modulation depth, which also agreed well with the theoretical prediction in Fig. 10. Again, comparing the standard deviations of different modulation schemes with the same modulation depth suggested that the rate detected with SSMB was the most accurate at any given modulation depth, followed by FSM and SWM. This result can be reasonably attributed to the crosstalk reduction effect of the SSMB.

To verify the elimination of the crosstalk with SSMB at the modulation depth of $2\pi/3$, an improvement of the ARW at this bias point is needed, and this issue can be addressed by using the RIN suppressing technique, which is currently under study.

4. Conclusion

We have proposed and demonstrated two types of SSM schemes to suppress the crosstalk effects in an IFOG: SSMA and SSMB. The basic suppression principle is to reduce the correlation between the modulation voltage and the demodulation reference by separating their fundamental frequencies using the SSM schemes. The crosstalk can be further reduced or eliminated by adjusting the modulation depth. In an experimental IFOG with a 1500-m fiber coil, the measured ARW and bias instability performance using SSMB were better than those using FSM, SWM, and SSMA. Moreover, there was no evidence of dead band using SSMB at the input rate resolution of $0.002\text{ }^{\circ}/\text{h}$, whereas there was approximately $0.08\text{ }^{\circ}/\text{h}$ dead band using SWM and approximately $0.02\text{ }^{\circ}/\text{h}$ dead band using FSM. Although the SSMA was demonstrated to be impractical because of its high noise performance, the SSMB proved to be an appealing modulation scheme with improved IFOG performance.

The most attractive advantage of SSMB is that it offers the possibility of completely eliminating the crosstalk-induced bias error at the modulation depth of $2\pi/3$. To the best of our knowledge, this is the first time that a modulation scheme has exhibited this unique feature. By using the RIN suppressing technique, the optimal modulation depth can be modified at $2\pi/3$ (which is currently under study), leading to a high-performance IFOG with low noise, high sensitivity, wide dynamic range, and no dead band.

Funding

National Natural Science Foundation of China (NSFC) (61575012, 61575013).